

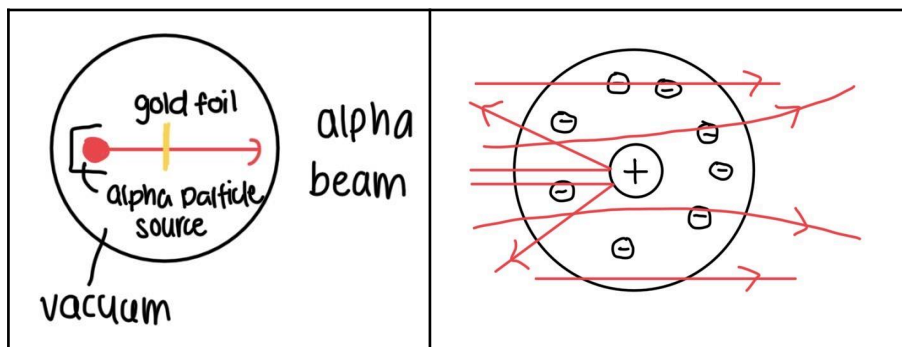
5.1 The nuclear model of the atom

Structure of atom:

- A very small nucleus surrounded by mostly empty space
- Nucleus is dense, contains most of the mass of the atom
- Nucleus is positively charged and negatively charged electrons are in orbit around the nucleus

Rutherford's experiment:

- During this experiment, a beam of alpha particles is directed at a thin gold foil suspended in a vacuum
- A vacuum is used, to prevent alpha particles from colliding with air particles



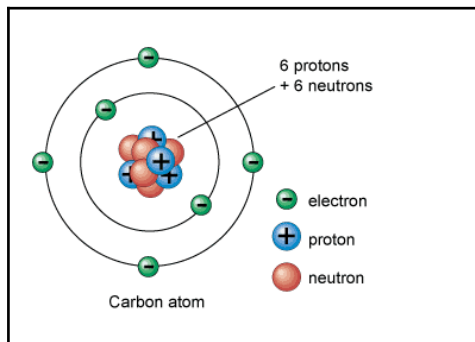
What happened - linked to structure

- Most alpha particles go straight through
- Atom has a nucleus surrounded by mostly empty space
- Some alpha particles are deflected at small angles
- Alpha particles are positive, and are reflected by a positively charged nucleus
- A small number of alpha particles are deflected back
- Nucleus is very small
- Nucleus is dense, contains most of the mass of the atom

Ions:

- When atoms lose electrons -> become positively charged ions
- When atoms gain electrons -> become negatively charged ions

Structure of the atom:

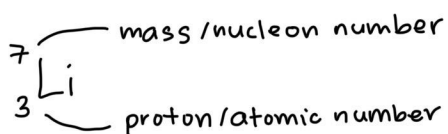


	relative charge	relative mass
Proton	+1	1
Neutron	0	1
Electron	-1	1/1840

Proton number (atomic number): the number of protons found in the nucleus of an atom

Nucleon number (mass number): the number of protons and neutrons found in the nucleus of an atom

Nuclide notation:



Isotopes:

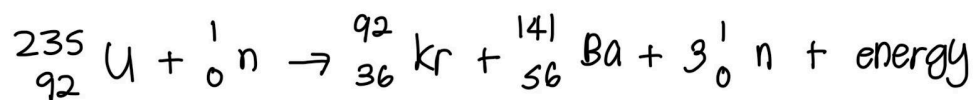
- **Isotope**: different atoms of the same element that have the same number of protons, but different number of neutrons
 - Isotopes tend to be more unstable and are more likely to decay, due to their imbalance of protons and neutrons
 - Unstable isotopes are the radioactive material that is found in the environment, which causes the background radiation
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Nuclear fission

Nuclear fission: the splitting of a large unstable nucleus into two smaller nuclei

- Nuclear fission takes place at nuclear power stations
- Reactant: unstable nucleus and 1 neutron
- Products: 2 daughter nuclei, 3 neutrons and energy

Nuclide equation:



- A neutron is absorbed by unstable nucleus, uranium-235
- The nucleus becomes highly unstable, uranium-236
- It splits into 2 lighter nuclei
- 3 neutrons and energy is released

More about the reaction:

- Mass of the products (2 daughter nuclei and 3 neutrons) is slightly less than the mass of the original nucleus, because the lost mass is converted to energy

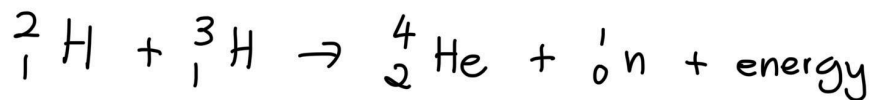
- It is a chain reaction - because, the emitted neutrons go on to split other nuclei, and nuclear fission takes place again
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Nuclear fusion

Nuclear fusion: the combining of two light nuclei to form a heavier nucleus

- Nuclear fusion is used to create energy in the stars and sun -> this is why they shine brightly
- Reactants: 2 parent nuclei
- Products: 1 daughter nucleus, 1 neutron and energy

Nuclide equation:



- 2 light nuclei fuse together
- A heavier daughter nucleus, 1 neutron and energy is released

More about the reaction:

- Mass of the products (daughter nucleus and 1 neutron) is less than the mass of the original nuclei, because the lost mass is converted to energy
 - It is not a chain reaction - this is because it does not require any of the products emitted as a reactant in the reaction.
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5.2 Radioactivity

Background radiation: radiation always present in the environment

Sources of background radiation:

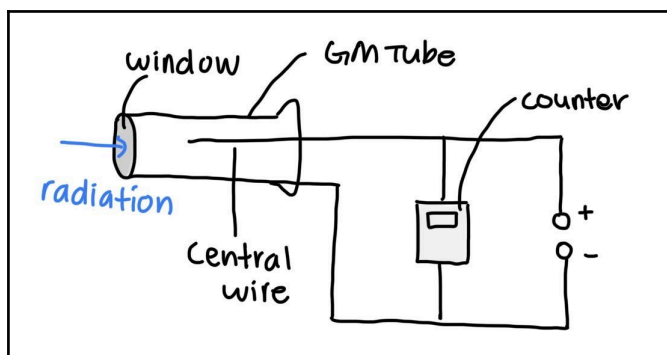
- radon gas
 - rocks
 - buildings
 - food and drink
 - cosmic rays
-

Ionising radiation:

- When unstable nuclei decay, they give out ionising radiation
 - Ionising radiation causes atoms to gain or lose electrons, forming ions
 - In living things, ionisation can damage or destroy cells
 - If a gas becomes ionised, it will conduct an electric current
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- Ionising radiation: alpha particles, beta particles and Gamma rays
 - Other forms of ionising radiation: Ultraviolet and X rays
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Detection of radiation:

- Measured using a Geiger muller tube (GM tube) connected to a counter
- It is in count rate - the number of decays per unit time
- Measured in counts/min or counts/s



- Window allows all radiation to pass through
- Radiation enters the tube, and ionises the gas inside

- This sets off a pulse of current in the circuit, which causes the counter to record it as 1 count

Note -> when radiation from a radioactive source is measured, the reading always includes background radiation.

- So, you need to minus the background radiation to find the actual value

The 3 types of nuclear emission

Alpha particles:

- Symbol: ${}^4_2\text{He}$ or 4_2a or ${}^4_2\alpha$
- Nature of radiation: helium nucleus, containing 2 protons and 2 neutrons
- Relative charge: +2
- Strong ionising effect -> low kinetic energy and high charge
- Low penetrating power (stopped by a sheet of paper)

Beta particles:

- Symbol: ${}^0_{-1}\beta$ or ${}^0_{-1}e$
- Nature of radiation: high speed electron emitted from the nucleus of an atom
- Relative charge: -1
- Weaker ionising effect than alpha, but stronger than gamma -> higher kinetic energy and lower charge than alpha
- Higher penetrating power than alpha, but lower than gamma (it is stopped by a few millimetres of aluminium foil)

Gamma rays:

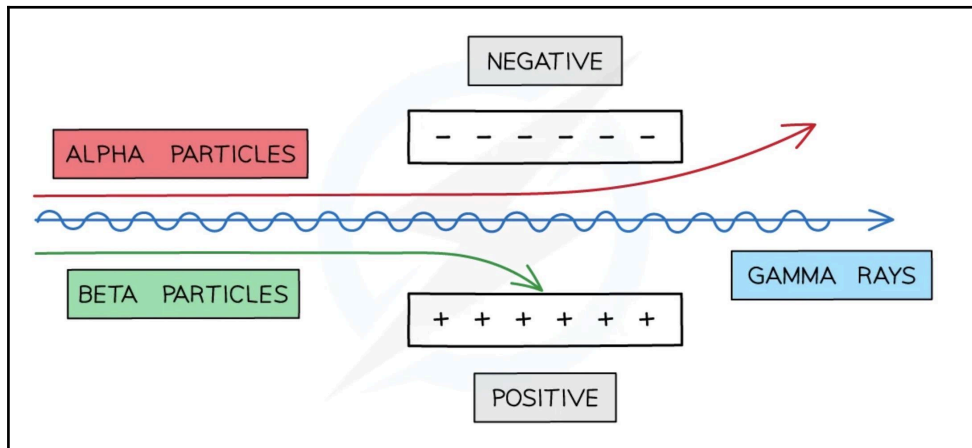
- Symbol: ${}^0_0\gamma$

- Nature of radiation: electromagnetic wave emitted from the nucleus of an atom
- Relative charge: 0
- Weakest ionising effect -> high kinetic energy and low charge
- Highest penetrating power (stopped by a few centimetres of lead)

Other things to note:

- relative mass -> $\alpha > \beta > \gamma$
- speed (kinetic energy) -> $\gamma > \beta > \alpha$

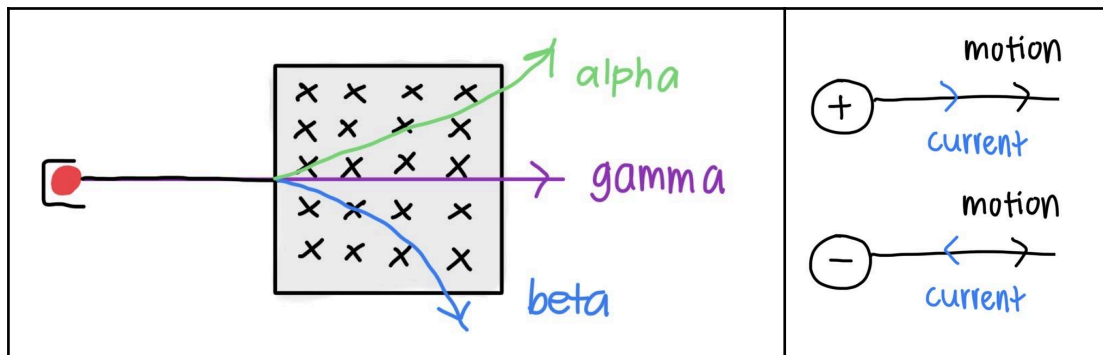
Effects of nuclear radiation in an electric field:



Explanation:

- Alpha particles deflect towards the negative plate, because they have a positive charge, and are attracted to the negative plate
- Alpha particles deflect less compared to beta particles, because it has a heavier mass
- Beta particles deflect towards the positive plate, because they have a negative charge, and are attracted to the positive plate
- Beta particles deflect more compared to Alpha particles, because it has a lighter mass
- Gamma ray is not deflected, because they have no charge

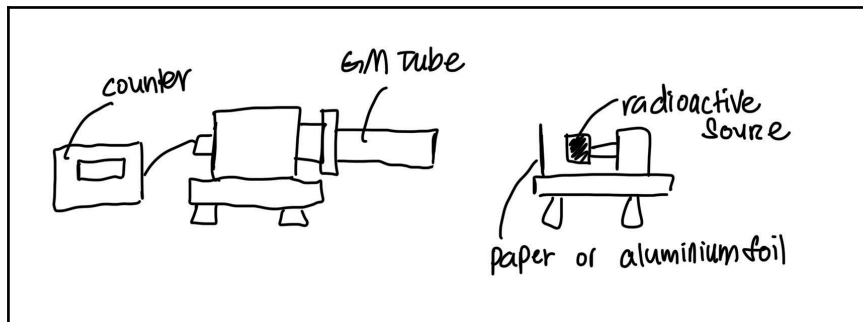
Effects of nuclear radiation in a magnetic field:



Explanation:

- Positively charged alpha particle creates an electric current in the direction of its motion
- The current creates a magnetic field that interacts with the uniform magnetic field, inducing a force that acts on the alpha particle upwards
- Force is always perpendicular to current, causing alpha particle to move in a circular path upwards
- Alpha particle deflects less than beta particle, because it has a heavier mass
- Negatively charged beta particle creates an electric current in the opposite direction of its motion
- The current creates a magnetic field that interacts with the uniform magnetic field, inducing a force that acts on the beta particle downwards
- Force is always perpendicular to current, causing beta particle to move in a circular path downwards
- Beta particle deflects more than Alpha particle because it has a lighter mass
- Gamma ray is not deflected, because it has no charge, and does not create an electric current

When determining the amount of radiation emitted from a radioactive source



Example - background radiation = 20 counts/min, average count rate with radioactive source = 560 counts/min, so actual = $560 - 20 = 540$ counts/min

When you place paper in front of the radioactive source:

- If average count rate = 20 counts/min $\rightarrow$ this is equal to background radiation, so only alpha is emitted
- If average count rate = 250 counts/min $\rightarrow$ 230 counts/min are beta and gamma, and 310 counts/min are alpha
- If average count rate = 560 counts/min $\rightarrow$ 540 counts/min are beta and gamma, no alpha is emitted

When you place aluminium foil in front of the radioactive source:

- If average count rate = 20 counts/min $\rightarrow$ this is equal to background radiation, only alpha and beta are emitted
- If average count rate = 250 counts/min $\rightarrow$ 230 counts/min are gamma while 310 counts/min are alpha and beta
- If average count rate = 560 counts/min, 540 counts/min are gamma, only gamma is emitted

Radioactive decay

- The most stable nuclei have the same number of protons and neutrons
- If a nucleus has an imbalance of protons and neutrons, it is more likely to decay into smaller nuclei until it gets to a stable nucleus with roughly the same number of each

Isotopes of an element may be radioactive because:

- There is an excess of protons or neutrons in the nucleus
- The nucleus is too heavy

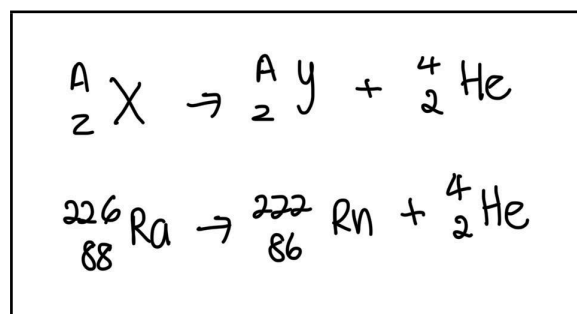
Radioactive decay: a change in an unstable nucleus, that can result in the emission of alpha particles, beta particles or gamma radiation

- Radioactive decay is a random process - cannot predict when a specific nucleus will decay
 - But, we know how long it takes for them to decay completely
- Radioactive decay is spontaneous - it occurs without any external influence
 - factors like temperature and pressure do not affect the decay itself

Note -> the value of background radiation may vary over time, due to the random nature of radioactive decay

Alpha decay:

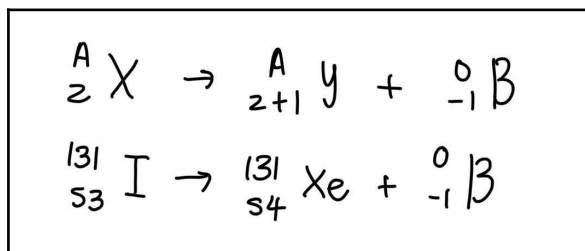
- Heavy unstable nucleus breaks apart to emit an alpha particle



- The number of protons decrease by 2
- The number of neutrons decrease by 2
- The number of of nucleons decrease by 4

Beta decay:

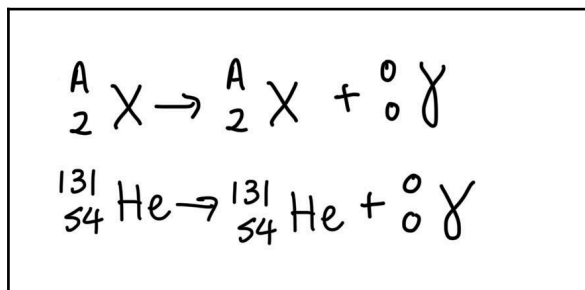
- An unstable nucleus changes a neutron into a proton, and then emits an electron



- The number of protons increase by 1
- The number one neutrons decrease by 1
- The number of nucleons remain the same

Gamma decay:

- An unstable nucleus with excess energy emits gamma rays to reduce its energy
- does not change number of protons, neutrons or nucleons
- Same element, but with lower energy



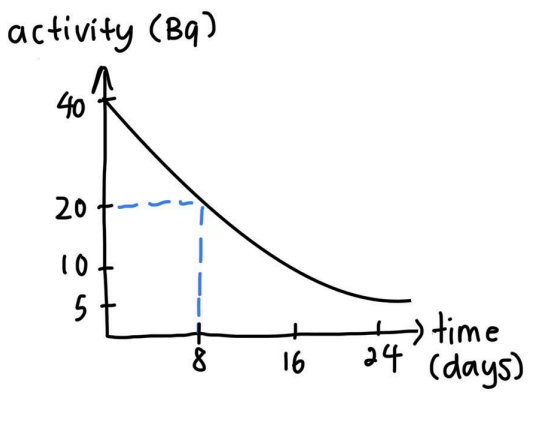
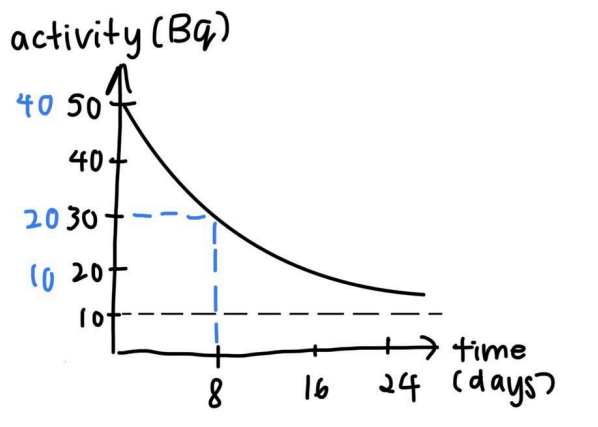
So, in conclusion:

- During alpha and beta decay, the nucleus changes to that of a different element
- During gamma decay, it is the same element but with lower energy

Half-life

Half-life of a particular isotope: the time taken for half of the nuclei of that isotope in any sample to decay

- Activity of radioactive sample is measured in becquerel (Bq)
- Count rate is proportional to this, but not equal because not all emitted radiation is detected by GM tube
- Half life is the time taken for the activity of any sample to decrease to half of its original value

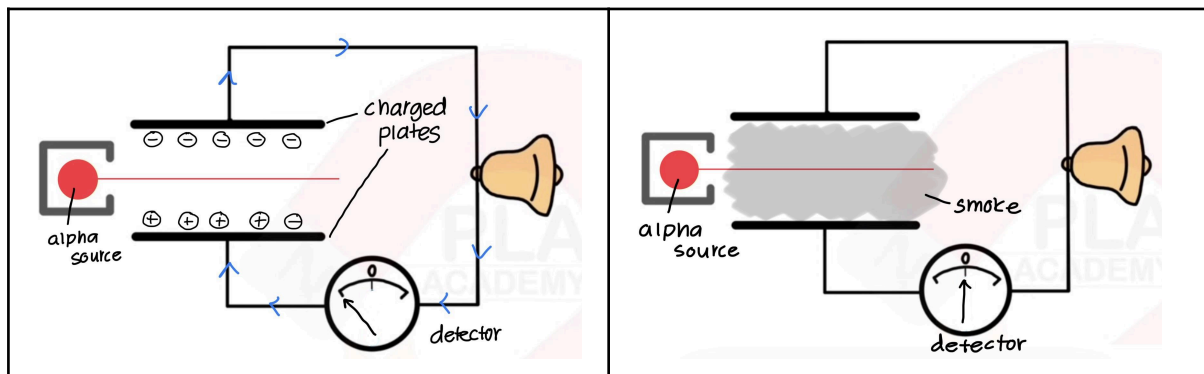
	
- $40 / 2 = 20\text{Bq}$ - So, one half life: $40 \rightarrow 20$ - Activity halves every 8 days - So, half-life is 8 days	- Background radiation = 10Bq - Initial activity = 50Bq - Actual radiation = $50 - 10 = 40\text{Bq}$ - $40 / 2 = 20\text{Bq}$ - So, one half life: $40 \rightarrow 20$ - With background, $20 + 10 = 30\text{Bq}$ - One half-life is 8 days

Note -> the graph may include background radiation, so you need to account for this when calculating half-life

- Certain isotopes have a short half life, so that after some time, little radiation will be emitted.
- Isotopes with a short half life can be hazardous, because a large amount of radiation is emitted in a short time.

Uses of radiation

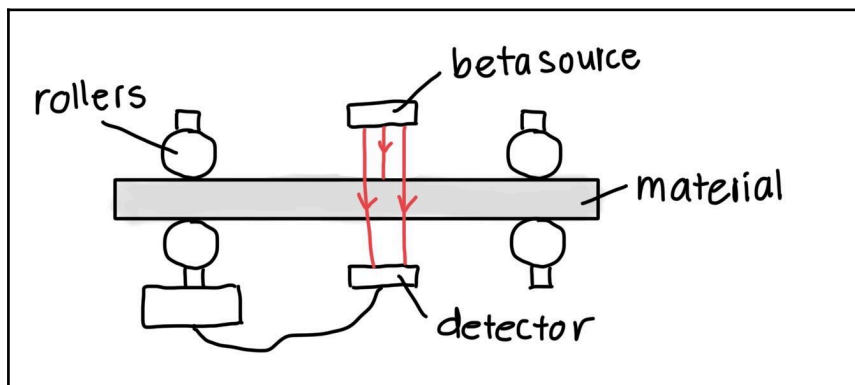
Smoke detectors: (Alpha radiation)



- Alpha source emits alpha radiation that passes through air between charged plates
- Alpha radiation ionises air particles, forming positively charged ions and negatively charged electrons
- These charges are attracted to the charged plates, causing the current to flow
- The detector senses this current.
- When smoke enters the detector, it absorbs the alpha radiation.
- This reduces ionisation, causing current to decrease
- When current drops below a certain level, the detector triggers the alarm

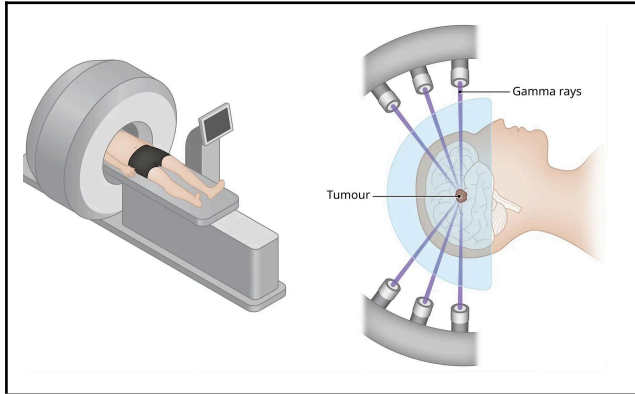
- Alpha radiation is used, because it is able to be absorbed by the smoke, ensuring that the alarm is triggered.
 - If beta or gamma radiation was used, they would pass straight through the smoke, and the alarm would not be triggered.
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Measuring thickness of materials: (Beta radiation)



- Particles that are able to penetrate through the material are detected by a detector
 - If material gets thicker, more particles will be absorbed by the material and less will penetrate through the material
 - This triggers the rollers to apply more pressure
 - If material gets thinner, more particles will penetrate through the material, and less will be absorbed by the material
 - This triggers the rollers to apply less pressure
- Beta radiation is used, because it will be partially absorbed by the material
- If Alpha radiation were used, all would be absorbed, and none would penetrate
- If Gamma radiation was used, all would penetrate, and none would be absorbed

Diagnosis and treatment of cancer (Gamma rays)



- Radiation can kill living cells
- Some cells, such as bacteria and cancer cells are more prone to radiation than others
- Beams of gamma rays are directed at the cancerous tumour
- Gamma rays are used because they are able to penetrate through the body and reach the tumour
- Alpha and beta particles are not used, as they would be absorbed by the body.

Sterilising medical equipment: (Gamma radiation)

- It is used to kill microorganisms on the medical equipment
- Items to be sterilised are placed in secure bags, to ensure they are not re-contaminated before use
- Gamma rays will pass through packaging and destroy bacteria, without damaging item

Irradiating food to kill bacteria: (Gamma radiation)

- It is used to kill microorganisms that are present on the food

- This makes the food last longer
 - This also reduces the risk of food-borne infections
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Effects and Safety Precautions

Effects of ionising radiation:

- Damage to cells and tissues
- Mutation
- Cancer

Safety Precautions:

- | |
|--|
| <ol style="list-style-type: none">1. Reduce exposure time2. Increase distance between source and living tissue3. Use shielding to absorb radiation |
|--|

- Store radioactive sources in lead boxes, and keep at a distance from people
 - Handle radioactive materials only while wearing gloves and using tongs, to increase distance from them
 - Wear lead plated suit to prevent body contamination
 - Limit the amount of time you handle sources for
 - Return the material to lead boxes, as soon as you finish using them
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